

Chapter 14B

Female Reproductive Disorders, Infections, Screening, and Malignancy

Medvale Internal Medicine - Reproductive Medicine Module

Chapter purpose

Chapter 14A built the endocrine and symptom framework. Chapter 14B turns that framework into disease-level reasoning: benign gynecologic disorders, contraception and pregnancy-aware medicine, infectious syndromes, cancer screening, and female reproductive malignancy. The wording intentionally emphasizes cause-effect links, diagnostic forks, and treatment rationales for downstream Graph-RAG extraction.

Major organizing rule: female reproductive disease is rarely isolated to one domain. PCOS is endocrine plus metabolic plus endometrial risk. PID is infection plus infertility risk. Contraception is reproductive planning plus thrombosis risk plus chronic disease compatibility. Postmenopausal bleeding is a symptom, but its most important relationship is to endometrial cancer exclusion.

14B.1 High-Yield Disease Map

The practical diagnostic map begins by sorting female reproductive disease into five overlapping categories: endocrine-anovulatory disease, structural uterine or pelvic disease, infectious/inflammatory disease, contraception or pregnancy-related decision-making, and malignancy or premalignancy. A patient can sit in more than one category. For example, chronic anovulation from PCOS can present as irregular bleeding, infertility, hirsutism, metabolic syndrome, and endometrial hyperplasia risk. Endometriosis can present as cyclic pelvic pain, dyspareunia, dyschezia, and infertility without a visible abnormality on routine pelvic examination.

The most important cognitive error is to label a reproductive symptom as benign before excluding dangerous mimics. Pelvic pain can be ectopic pregnancy, ovarian torsion, PID, tubo-ovarian abscess, appendicitis, nephrolithiasis, or inflammatory bowel disease. Abnormal uterine bleeding can be medication-related, anovulatory, structural, pregnancy-related, or malignant. Vaginal discharge can be vaginitis, cervicitis, foreign body, malignancy, physiologic leukorrhea, or genitourinary syndrome of menopause.

Clinical entry point	Common benign causes	Dangerous or high-consequence diagnoses to exclude	First reasoning step
Irregular menses / AUB	PCOS, fibroids, adenomyosis, anovulation, iatrogenic bleeding	Pregnancy complication, endometrial hyperplasia/cancer, coagulopathy	Pregnancy test first; then age/risk-based AUB workup
Chronic pelvic pain	Endometriosis, adenomyosis, pelvic floor dysfunction, IBS, bladder pain syndrome	Ovarian cancer, chronic PID, inflammatory bowel disease, pelvic mass	Separate cyclic pain, infection signs, mass symptoms, urinary/GI features
Acute pelvic pain	Ruptured ovarian cyst, dysmenorrhea, PID	Ectopic pregnancy, torsion, TOA, appendicitis	Assess pregnancy status, hemodynamics, fever, peritoneal signs
Vaginal discharge	BV, candidiasis, trichomoniasis, physiologic discharge	Cervicitis, PID, STI complications, retained foreign body	Distinguish vaginitis from cervicitis/PID
Postmenopausal bleeding	Atrophy, polyps, anticoagulation	Endometrial cancer until proven otherwise	Do not reassure before endometrial evaluation
Breast lump	Cyst, fibroadenoma, fat necrosis	Breast cancer, inflammatory breast cancer	Triple assessment: exam, imaging, tissue when indicated

14B.2 Polycystic Ovary Syndrome

Polycystic ovary syndrome is a chronic anovulatory and hyperandrogenic disorder with reproductive, dermatologic, metabolic, and endometrial consequences. The name is imperfect because the syndrome is not defined by ovarian cysts alone. The core physiology is increased androgen exposure and disordered follicular maturation. In many patients, insulin resistance amplifies ovarian androgen production and suppresses hepatic sex hormone-binding globulin, increasing free testosterone. Increased androgens disrupt normal follicle selection, producing oligo-ovulation or anovulation. Anovulation causes irregular menses and infertility; chronic anovulation also produces prolonged unopposed estrogen exposure, which increases endometrial hyperplasia and endometrial cancer risk.

Graph relationship chain

Insulin resistance -> hyperinsulinemia -> increased ovarian androgen production and decreased SHBG -> increased free testosterone -> hyperandrogenic symptoms and follicular dysfunction -> oligo-ovulation/anovulation -> irregular bleeding and infertility. Anovulation -> absent progesterone withdrawal -> unopposed estrogen -> endometrial proliferation -> hyperplasia/cancer risk.

Adult PCOS is usually diagnosed clinically using Rotterdam-style criteria after excluding mimics. The patient typically needs two of three features: oligo- or anovulation, clinical or biochemical hyperandrogenism, and polycystic ovarian morphology. In adolescents, diagnosis is more cautious because physiologic pubertal irregularity and acne can mimic PCOS. Do not diagnose PCOS simply because an ultrasound report says polycystic ovaries; polycystic morphology can be seen in people without the syndrome.

PCOS domain	What it means clinically	Common evidence	Important mimics to exclude
Ovulatory dysfunction	Oligomenorrhea, amenorrhea, irregular heavy bleeding, infertility	Cycles often >35 days, fewer than 8-9 menses/year, unpredictable bleeding	Pregnancy, thyroid disease, hyperprolactinemia, hypothalamic amenorrhea, primary ovarian insufficiency
Hyperandrogenism	Hirsutism, acne, androgenic alopecia; severe virilization is atypical	Elevated total or free testosterone; clinical hirsutism	Nonclassic CAH, androgen-secreting tumor, Cushing syndrome, medication exposure

PCOS domain	What it means clinically	Common evidence	Important mimics to exclude
Polycystic ovarian morphology	Many small follicles and increased ovarian volume by ultrasound criteria	Ultrasound when needed, not required if other two criteria present	Normal variant, adolescent physiology, ovarian hyperstimulation context
Metabolic phenotype	Insulin resistance, obesity, dyslipidemia, impaired glucose tolerance, OSA risk	A1c/glucose assessment, lipid screening, BP, BMI/waist risk	Type 2 diabetes, metabolic syndrome not due to PCOS alone

Treatment logic in PCOS

Treatment depends on the patient goal. If the goal is cycle control and endometrial protection, combined hormonal contraception is often first-line unless contraindicated; cyclic progestin or a levonorgestrel IUD can also protect the endometrium. If hirsutism is the dominant symptom, combined hormonal contraception lowers ovarian androgen production and increases SHBG; spironolactone can be added but requires reliable contraception because antiandrogen exposure can affect fetal sexual development. If pregnancy is desired, ovulation induction is the reproductive target. If metabolic risk is dominant, weight management, diabetes screening, lipid management, blood pressure control, and metformin in selected patients are important.

PCOS should not be treated as a cosmetic diagnosis. The long-term internal medicine problem is that an endocrine reproductive syndrome can signal cardiometabolic risk and endometrial risk. A patient with PCOS needs evaluation for diabetes risk, obstructive sleep apnea risk when clinically suggested, dyslipidemia, hypertension, mood symptoms, and endometrial protection if prolonged amenorrhea or infrequent bleeding persists.

14B.3 Endometriosis and Adenomyosis

Endometriosis is the presence of endometrial-like glands and stroma outside the uterine cavity. Lesions respond to hormonal cycling and create inflammation, fibrosis, adhesions, and pain sensitization. Classic symptoms include dysmenorrhea, chronic pelvic pain, deep dyspareunia, dyschezia, dysuria, and infertility. The pain pattern may be cyclic, but advanced disease can cause noncyclic pain because adhesions and neural sensitization persist even outside menses.

The diagnosis is often clinical. Laparoscopy with histology can confirm disease, but many patients are treated empirically when the presentation is typical and no red flags suggest malignancy, pregnancy complication, or acute surgical pathology. A normal ultrasound does not rule out endometriosis because superficial peritoneal implants may be invisible. Ultrasound is useful for endometriomas, adnexal masses, and alternative structural disease.

Adenomyosis is endometrial tissue within the myometrium. It usually produces a bulky, tender uterus with heavy menstrual bleeding and dysmenorrhea. Adenomyosis overlaps clinically with fibroids and endometriosis, but the anatomic lesion is different: adenomyosis is within the uterine muscle, fibroids are smooth muscle tumors, and endometriosis is outside the uterine cavity.

Feature	Endometriosis	Adenomyosis	Fibroids
Core lesion	Endometrial-like tissue outside uterus	Endometrial tissue within myometrium	Benign monoclonal smooth muscle tumors
Pain pattern	Cyclic pelvic pain, dysmenorrhea, deep dyspareunia, dyschezia; can become chronic	Dysmenorrhea, uterine tenderness, heavy bleeding	Pressure, bulk symptoms, heavy bleeding; pain if degeneration or large mass
Fertility link	Inflammation, adhesions, endometriomas, altered pelvic anatomy	Can impair fertility but less classic than endometriosis	Submucosal/intracavitary distortion strongly affects fertility
Exam/imaging clue	May be normal exam; endometrioma on US; laparoscopy confirms	Globular enlarged tender uterus; MRI/US may suggest	Enlarged irregular uterus; ultrasound identifies number/location
Treatment logic	NSAIDs, hormonal suppression, progestins, GnRH analogs/antagonists, surgery for refractory disease or fertility goals	NSAIDs, hormonal suppression, LNG-IUD, hysterectomy if definitive and no fertility desire	Medical bleeding control, myomectomy, uterine artery embolization, hysterectomy depending goals

Why endometriosis causes infertility

Endometriosis -> chronic pelvic inflammation -> adhesions and distorted tubo-ovarian anatomy -> impaired oocyte pickup and transport. Endometriomas and ovarian surgery can also reduce ovarian reserve. Thus endometriosis links pelvic pain and infertility through both inflammatory and mechanical pathways.

14B.4 Uterine Leiomyomas, Polyps, and Structural Bleeding

Uterine leiomyomas, or fibroids, are benign smooth muscle tumors of the myometrium. Their clinical effects depend more on location than size alone. Submucosal fibroids distort the endometrial cavity and are strongly associated with heavy menstrual bleeding and infertility. Intramural fibroids can cause bleeding, pain, and reproductive effects when large or cavity-distorting. Subserosal fibroids often cause bulk symptoms such as pelvic pressure, urinary frequency, constipation, or abdominal distension.

Endometrial polyps are focal overgrowths of endometrial glands and stroma. They commonly cause intermenstrual bleeding, heavy bleeding, or postmenopausal bleeding. Most are benign, but malignancy risk rises with age, postmenopausal status, abnormal bleeding, tamoxifen exposure, and other endometrial cancer risk factors. Because polyps and fibroids can both cause bleeding, the diagnostic question is whether the endometrial cavity is distorted or contains a focal lesion. Saline infusion sonography and hysteroscopy are useful when routine ultrasound does not fully characterize the cavity.

Structural entity	Typical symptoms	Fertility implications	Diagnostic tools	Management concept
Submucosal fibroid	Heavy bleeding, cramping, anemia	High impact due to cavity distortion	TVUS, saline sonography, hysteroscopy	Hysteroscopic myomectomy when symptomatic or fertility goal
Intramural fibroid	Heavy bleeding, pain, enlarged uterus	Impact depends on size and cavity distortion	TVUS/MRI when mapping needed	Medical bleeding control, myomectomy, UAE, hysterectomy by goals
Subserosal fibroid	Bulk symptoms, pressure, urinary frequency	Less fertility impact unless very large/distorting	TVUS; MRI for surgical planning	Observation or procedure if symptomatic

Structural entity	Typical symptoms	Fertility implications	Diagnostic tools	Management concept
Endometrial polyp	Intermenstrual bleeding, heavy bleeding, postmenopausal bleeding	Can impair implantation	TVUS, saline sonography, hysteroscopy	Polypectomy when symptomatic, postmenopausal, or risk factors

Abnormal bleeding should be connected to systemic consequences. Heavy menstrual bleeding can cause iron deficiency anemia; anemia can cause fatigue, dyspnea, palpitations, restless legs, and reduced exercise tolerance. Therefore, reproductive bleeding is not only a gynecologic symptom; it is also a hematologic and cardiovascular stressor.

14B.5 Menopause, Genitourinary Syndrome of Menopause, and Hormone Therapy Logic

Menopause is diagnosed retrospectively after 12 months without menses due to loss of ovarian follicular function. The transition is characterized by fluctuating estrogen, irregular ovulation, vasomotor symptoms, sleep disturbance, mood changes, and eventually low-estrogen genitourinary and bone effects. Menopause is physiologic, but its symptoms can be clinically significant and its low-estrogen state changes risk profiles for osteoporosis, genitourinary symptoms, and cardiometabolic health.

Genitourinary syndrome of menopause results from estrogen deficiency in the vulva, vagina, urethra, and bladder. It can cause vaginal dryness, burning, dyspareunia, recurrent urinary symptoms, urinary urgency, and recurrent UTIs. Unlike hot flashes, GSM often persists or worsens without treatment. Local vaginal estrogen, moisturizers, lubricants, and other local therapies are often more targeted than systemic hormone therapy for isolated GSM.

Systemic menopausal hormone therapy is most useful for bothersome vasomotor symptoms in appropriately selected patients. The risk-benefit profile depends on age, time since menopause, uterus status, cardiovascular risk, breast cancer risk, thromboembolic risk, and symptom burden. Estrogen alone can be used only if the patient lacks a uterus; otherwise, unopposed systemic estrogen increases endometrial hyperplasia and cancer risk, so a progestogen is required for endometrial protection.

Clinical issue	Estrogen relationship	Treatment logic	Do not miss
Vasomotor symptoms	Thermoregulatory instability linked to estrogen withdrawal	Systemic HT if appropriate; nonhormonal options if contraindications	Breast cancer history, VTE/stroke risk, CAD risk, liver disease
GSM	Low estrogen causes atrophic vulvovaginal and urinary changes	Lubricants/moisturizers; local vaginal estrogen if persistent	Persistent bleeding, lesion, or mass needs evaluation
Uterus present	Unopposed estrogen stimulates endometrium	Add progestogen if systemic estrogen used	Postmenopausal bleeding is never normal until evaluated
Bone health	Estrogen loss accelerates bone resorption	Calcium/vitamin D adequacy, weight-bearing exercise, DEXA risk-based, osteoporosis meds when indicated	Fragility fracture risk

14B.6 Contraception and Reproductive Planning

Contraception should be taught as a risk-stratified medical intervention, not just as pregnancy prevention. Each method has a mechanism, effectiveness profile, adverse effect pattern, contraindication set, and noncontraceptive benefits. For internal medicine, the most important task is matching the method to the patient: thrombosis risk, migraine with aura, hypertension, smoking age over 35, diabetes complications, obesity, postpartum state, liver disease, breast cancer history, medications, and reproductive goals all matter.

The major method categories are combined estrogen-progestin methods, progestin-only methods, intrauterine devices, barrier methods, emergency contraception, and sterilization. Long-acting reversible contraception includes IUDs and implants and has high effectiveness because it does not rely on daily adherence. Combined hormonal contraception can reduce dysmenorrhea, heavy menstrual bleeding, acne, endometriosis pain, and ovarian/endometrial cancer risk, but estrogen exposure can increase venous thromboembolism risk and is contraindicated in selected high-risk states.

Method	Mechanism	Major benefits	Major concerns / contraindication logic
Combined pill/patch/ring	Suppresses ovulation; thickens cervical mucus; stabilizes endometrium	Cycle control, acne improvement, dysmenorrhea reduction, less bleeding, ovarian/endometrial cancer risk reduction	Avoid estrogen in migraine with aura, prior VTE/stroke, severe HTN, smoking age >35 with heavy use, high-risk thrombophilia, active breast cancer
Progestin-only pill	Thickens cervical mucus; some formulations suppress ovulation	Estrogen-free option; useful when estrogen contraindicated	Adherence sensitivity for some pills; irregular bleeding
Depot medroxyprogesterone	Suppresses ovulation	Very effective, private, estrogen-free; reduces endometriosis pain	Weight gain, irregular bleeding, delayed return to fertility, bone density concern with prolonged use
Etonogestrel implant	Suppresses ovulation and thickens mucus	Highly effective LARC; rapid return to fertility	Unpredictable bleeding is common
Levonorgestrel IUD	Local progestin effect: thickened mucus and endometrial suppression	Very effective; heavy bleeding reduction; endometrial protection	Insertion pain/perforation rare; expulsion; initial spotting
Copper IUD	Spermicidal inflammatory copper effect	Hormone-free; effective emergency contraception when placed promptly	Heavier bleeding/cramps can worsen baseline menorrhagia
Barrier methods	Blocks sperm exposure; condoms reduce STI transmission	STI protection with condoms; nonhormonal	Higher typical-use failure than LARC; user-dependent

Estrogen contraindication pattern

Estrogen-containing contraception should trigger a vascular risk check. Estrogen + high-risk thrombosis state -> increased VTE/stroke concern. Migraine with aura, prior thromboembolism, stroke, uncontrolled hypertension, and smoking after age 35 are classic red flags. When estrogen is unsafe, consider progestin-only or nonhormonal methods.

Emergency contraception and missed opportunity prevention

Emergency contraception works before implantation and should be offered after unprotected intercourse, contraceptive failure, or sexual assault when pregnancy is not desired. Options include oral levonorgestrel, ulipristal acetate, and copper IUD placement. The copper IUD is the most effective emergency contraception and provides ongoing contraception. Ulipristal can be more effective than levonorgestrel closer to 5 days after intercourse, but follow-up contraception timing and pregnancy exclusion matter.

A useful Graph-RAG relation is: unprotected intercourse -> possible fertilization risk -> emergency contraception can reduce pregnancy risk before implantation. Emergency contraception does not treat STIs, does not terminate an established pregnancy, and does not replace evaluation after sexual assault or STI exposure.

Pregnancy-aware medicine

Pregnancy-aware medicine means that medication decisions, diagnostic imaging, preventive care, chronic disease management, and counseling should account for the possibility of pregnancy. Teratogenic medications should be identified before conception when possible. Examples relevant to internal medicine include isotretinoin, valproate, warfarin, ACE inhibitors/ARBs later in pregnancy, methotrexate, mycophenolate, and some antiseizure or immunosuppressive medications. The relationship is not simply medication -> fetal risk; it is medication indication -> maternal benefit vs fetal risk -> safer alternative if appropriate -> shared decision-making.

Medication / exposure	Pregnancy concern	Clinical planning implication
Isotretinoin	High teratogenic risk	Strict pregnancy prevention program and testing
Valproate	Neural tube and neurodevelopmental risk	Avoid when alternatives exist, especially for migraine or mood indications
Warfarin	Embryopathy and fetal bleeding risk	Use pregnancy-compatible anticoagulation when indicated
ACE inhibitor / ARB	Fetal renal injury especially later pregnancy	Switch before conception or when pregnancy recognized if possible
Methotrexate	Antifolate teratogenicity and pregnancy loss	Contraindicated in pregnancy except special contexts such as ectopic treatment
Mycophenolate	Pregnancy loss and congenital malformation risk	Use safer immunosuppression plan before conception

Pregnancy complications also predict future disease. Gestational diabetes increases later type 2 diabetes risk. Hypertensive disorders of pregnancy increase future chronic hypertension and cardiovascular risk. Preterm delivery, fetal growth restriction, and placental disease can also serve as vascular risk markers. Therefore, pregnancy history belongs in the adult internal medicine history, not only in obstetric records.

14B.7 Vaginitis: BV, Candidiasis, and Trichomoniasis

Vaginitis is inflammation or dysbiosis involving the vagina. The common triad is bacterial vaginosis, vulvovaginal candidiasis, and trichomoniasis. The first diagnostic split is whether symptoms localize to the vagina or cervix/upper genital tract. Vaginitis commonly causes discharge, odor, itching, burning, irritation, and dyspareunia. Cervicitis more often causes mucopurulent cervical discharge, friability, postcoital bleeding, or dysuria, and PID adds pelvic pain, cervical motion tenderness, uterine/adnexal tenderness, fever, or systemic symptoms.

Condition	Pathophysiology	Typical discharge / clues	pH / microscopy pattern	Treatment concept
Bacterial vaginosis	Loss of lactobacilli with anaerobic overgrowth	Thin gray-white discharge, fishy odor, worse after sex/menses	pH >4.5; clue cells; positive whiff test	Metronidazole or clindamycin regimens; recurrence common
Vulvovaginal candidiasis	Candida overgrowth, often <i>C. albicans</i>	Pruritus, burning, vulvar erythema; thick curdy discharge may occur	Usually normal pH; budding yeast/pseudohyphae on microscopy	Azole therapy or fluconazole; complicated/recurrent disease needs longer plan
Trichomoniasis	Sexually transmitted protozoan infection	Frothy yellow-green discharge, odor, irritation; strawberry cervix uncommon but classic	pH often >4.5; motile trichomonads if seen; NAAT more sensitive	Metronidazole/tinidazole; treat partners and avoid reinfection

Vaginitis diagnostic pitfall

A patient with discharge and pelvic pain should not be treated as simple vaginitis until cervicitis and PID are considered. Vaginitis affects the vaginal environment; cervicitis and PID imply ascending infection risk and reproductive tract complications.

14B.8 Cervicitis, PID, and Tubo-Ovarian Abscess

Cervicitis is inflammation of the cervix, commonly due to *Chlamydia trachomatis* or *Neisseria gonorrhoeae*, though other organisms can contribute. It can present with mucopurulent discharge, cervical friability, intermenstrual bleeding, postcoital bleeding, dysuria, or be asymptomatic. Untreated cervicitis can ascend to the upper genital tract, producing endometritis, salpingitis, PID, tubo-ovarian abscess, infertility, ectopic pregnancy, and chronic pelvic pain.

PID is a clinical diagnosis. Do not wait for perfect test confirmation when the presentation is compatible because delayed treatment increases reproductive harm. Minimal criteria include pelvic or lower abdominal pain with no better explanation plus cervical motion tenderness, uterine tenderness, or adnexal tenderness. Supportive features include fever, mucopurulent cervical discharge, elevated inflammatory markers, abundant WBCs on wet prep, positive gonorrhea/chlamydia testing, or imaging evidence of thickened fluid-filled tubes or tubo-ovarian complex.

Syndrome	Clinical clues	Complication pathway	Management principle
Cervicitis	Mucopurulent cervical discharge, friability, postcoital bleeding, dysuria; may be asymptomatic	Cervical infection -> ascending infection -> PID	NAAT for GC/CT; treat based on organism/risk and ensure partner management
PID	Pelvic pain plus CMT/uterine/adnexal tenderness; fever/discharge may occur	Endometritis/salpingitis -> scarring/adhesions -> infertility, ectopic pregnancy, chronic pain	Empiric broad coverage including GC/CT and anaerobes; low threshold to treat
Tubo-ovarian abscess	PID with adnexal mass, fever, severe pain, leukocytosis; imaging complex mass	Contained abscess -> rupture/sepsis risk; infertility risk	Hospitalization, IV antibiotics, drainage/surgery if rupture, sepsis, large/persistent abscess

When to hospitalize PID

Hospitalization is considered when surgical emergencies cannot be excluded, tubo-ovarian abscess is present, pregnancy is present, severe illness/nausea/vomiting/high fever prevents outpatient therapy, the patient cannot tolerate or follow outpatient therapy, or there is no clinical response to oral therapy. This decision links infection severity, diagnostic uncertainty, and reproductive risk.

Graph relationship chain

Cervicitis -> ascending infection -> PID -> tubal inflammation and scarring -> infertility + ectopic pregnancy risk + chronic pelvic pain.
PID + complex adnexal infection -> tubo-ovarian abscess -> rupture/sepsis risk.

14B.9 Genital Ulcers, HPV, and STI Prevention Links

Genital ulcers require syndrome-based reasoning. Painful grouped vesicles or shallow ulcers suggest HSV, whereas painless chancre with regional lymphadenopathy suggests primary syphilis. However, presentations overlap, coinfection occurs, and history/exam alone is imperfect. Testing should be directed by lesion type and epidemiologic risk: HSV PCR/NAAT from lesions when present, syphilis serology, HIV testing, and additional STI testing as appropriate.

Human papillomavirus causes anogenital warts through low-risk HPV types and cervical/vulvar/vaginal/anal/oropharyngeal cancers through persistent infection with high-risk types such as HPV 16 and 18. HPV vaccination prevents many HPV-associated cancers and warts, but vaccination does not treat established infection. Cervical screening is still needed in vaccinated patients because vaccination does not cover all oncogenic HPV types and many patients were exposed before vaccination.

Syndrome	Most likely causes	Key diagnostic move	Relationship to reproductive/cancer risk
Painful vesicles/ulcers	HSV-1 or HSV-2	Lesion PCR/NAAT; type-specific testing in selected contexts	Recurrent outbreaks, neonatal transmission risk if active near delivery
Painless ulcer	Syphilis until proven otherwise	Nontreponemal + treponemal serology; evaluate partners	Systemic disease if untreated; congenital infection risk
Exophytic warts	Low-risk HPV types, especially 6/11	Often clinical; biopsy if atypical, pigmented, fixed, ulcerated, or immunocompromised	Benign warts are distinct from high-risk HPV cancer pathway
Persistent high-risk HPV	HPV 16/18 and others	Pap/HPV screening and colposcopy pathways	Persistent infection -> dysplasia -> cervical cancer risk

14B.10 Cervical Cancer Screening and Cervical Dysplasia

Cervical cancer prevention rests on HPV vaccination, screening for precancer, and treatment of high-grade lesions before invasion. Persistent high-risk HPV infection is the causal driver for most cervical cancers. The transformation zone is the key anatomic site where squamous metaplasia and HPV-associated dysplasia arise, which is why sampling adequacy and colposcopic visualization matter.

Average-risk cervical screening starts at age 21 regardless of age at sexual debut. Ages 21-29 are generally screened with cytology alone every 3 years. Ages 30-65 can be screened with primary high-risk HPV testing every 5 years, cytology alone every 3 years, or cotesting every 5 years depending on accepted local guidelines and availability. Screening generally stops after 65 if the patient has had adequate prior negative screening and no high-risk history. Patients with immunosuppression, prior high-grade lesions, DES exposure, or cervical cancer history require individualized screening.

Age/risk group	Screening concept	Reasoning
<21 years	No routine cervical cancer screening	HPV infections and mild cytologic abnormalities often regress; screening increases harm without clear benefit
21-29 years	Cytology every 3 years for average risk	Cytology detects dysplasia while limiting overtreatment of transient HPV infection
30-65 years	HPV-based testing every 5 years, cotesting every 5 years, or cytology every 3 years	Persistent high-risk HPV is the major causal risk; HPV testing improves risk stratification
>65 years	Stop only if adequate prior negative screening and no high-risk history	Cancer risk remains if prior screening inadequate or high-grade history exists
High-risk patients	Individualized more intensive screening	Immunosuppression, prior CIN2+, DES exposure, or cervical cancer history change risk

Screening vs diagnosis

Screening tests estimate risk; they do not diagnose invasive cancer. Abnormal screening -> risk-based triage -> colposcopy/biopsy when indicated. Visible cervical mass, unexplained bleeding, or concerning symptoms require diagnostic evaluation regardless of screening interval.

Dysplasia language for concept mapping

CIN1 roughly corresponds to low-grade HPV effect and often regresses. CIN2 and CIN3 are high-grade precancerous lesions with higher progression risk, so excisional or ablative treatment is often considered depending on age, pregnancy status, fertility concerns, and risk estimates. The progression chain is persistent high-risk HPV -> cervical intraepithelial neoplasia -> invasive cervical cancer. The clinical goal is to interrupt the chain at the precancer stage.

14B.11 Endometrial Hyperplasia and Endometrial Cancer

Endometrial cancer is strongly linked to unopposed estrogen exposure. Estrogen stimulates endometrial proliferation, while progesterone differentiates and stabilizes the endometrium. When ovulation is absent, progesterone exposure is inadequate; the endometrium may proliferate continuously, producing hyperplasia and eventually malignancy. This is why PCOS, obesity, chronic anovulation, nulliparity, early menarche, late menopause, estrogen-only therapy with a uterus, and tamoxifen exposure are important risk factors.

Postmenopausal bleeding is endometrial cancer until proven otherwise. Premenopausal abnormal bleeding can also require endometrial sampling when age or risk factors raise concern. The risk-based logic is more important than memorizing a single age cutoff: abnormal bleeding plus unopposed estrogen risk, obesity, chronic anovulation, failed medical therapy, or persistent symptoms should prompt evaluation.

Risk factor	Mechanistic link to endometrium	Clinical implication
Obesity	Adipose aromatization increases estrogen exposure	Higher threshold for reassurance after AUB; counsel metabolic risk
PCOS/chronic anovulation	Low progesterone withdrawal; unopposed estrogen	Endometrial protection with cyclic progestin, CHC, or LNG-IUD when appropriate
Estrogen therapy with uterus	Estrogen stimulates proliferation without progestin opposition	Add progestogen if systemic estrogen and uterus present
Tamoxifen	Partial estrogen agonist effect in endometrium	Evaluate abnormal bleeding; do not screen asymptomatic routinely in average cases
Lynch syndrome	Mismatch repair defect increases endometrial and ovarian cancer risk	Genetic counseling, cascade testing, individualized surveillance/risk reduction

Core relationship

Anovulation -> insufficient progesterone -> unopposed estrogen -> endometrial proliferation -> hyperplasia -> endometrioid endometrial carcinoma risk. Postmenopausal bleeding -> diagnostic evaluation because bleeding after menopause is not physiologic.

14B.12 Ovarian Cancer and Adnexal Mass Reasoning

Ovarian cancer is dangerous because early symptoms are nonspecific and there is no recommended population screening test for average-risk asymptomatic patients. Symptoms can include bloating, early satiety, abdominal or pelvic pain, increased abdominal girth, urinary urgency/frequency, and unexplained weight change. The internal medicine clue is persistence and change from baseline, especially in a postmenopausal patient.

Adnexal mass evaluation integrates age, menopausal status, ultrasound morphology, symptoms, family history, and tumor markers when appropriate. Simple cysts in premenopausal patients are often functional and benign. Complex masses, solid components, papillary projections, ascites, bilateral masses, nodularity, or postmenopausal status increase concern. CA-125 is not a screening test for average-risk people; it is nonspecific and can rise with endometriosis, fibroids, PID, pregnancy, menstruation, liver disease, and many cancers.

Finding	Benign-leaning interpretation	Concerning interpretation
Premenopausal simple cyst	Often functional follicular or corpus luteum cyst	Persistent growth, large size, symptoms, complex features require follow-up
Postmenopausal adnexal mass	Some simple cysts still benign	Lower threshold for malignancy evaluation
Ultrasound morphology	Thin-walled, unilocular, simple fluid	Solid areas, papillary projections, septations, nodularity, ascites
CA-125	Can be elevated in many benign inflammatory states	More useful in postmenopausal risk stratification and treatment monitoring
Family history	No hereditary pattern lowers suspicion	BRCA/HBOC or Lynch pattern increases ovarian cancer risk

Hereditary ovarian cancer links

BRCA1 and BRCA2 mutations increase risk for breast and ovarian cancer and have implications for cascade testing, breast screening intensification, risk-reducing salpingo-oophorectomy timing, and family counseling. Lynch syndrome is more classically associated with colorectal and endometrial cancer but also increases ovarian cancer risk. A family history should be structured: age at diagnosis, lineage, bilateral disease, male breast cancer, ovarian/pancreatic/prostate cancers, colon/endometrial cancers, and known pathogenic variants.

14B.13 Breast Disease and Breast Cancer Screening

Breast symptoms are approached by age, pregnancy/lactation status, focality, associated skin/nipple findings, imaging results, and tissue diagnosis when needed. Fibroadenoma, cysts, mastalgia, duct ectasia, mastitis, abscess, fat necrosis, and malignancy can all present as breast complaints. The most important red flags are a hard fixed mass, skin dimpling, nipple retraction, spontaneous unilateral bloody nipple discharge, peau d orange, inflammatory erythema not responding as expected, axillary nodes, and systemic signs of malignancy.

Screening mammography is designed for asymptomatic average-risk patients. Current USPSTF guidance recommends biennial mammography for women aged 40-74. Screening differs from diagnostic evaluation: a patient with a palpable mass or concerning nipple/skin findings needs diagnostic imaging and often biopsy even if recently screened.

Breast condition	Typical presentation	Diagnostic/management concept
Fibroadenoma	Mobile, rubbery, well-circumscribed mass, often younger patient	Ultrasound/mammography by age; biopsy if atypical, enlarging, or uncertain
Simple cyst	Fluctuant or tender mass; may vary with cycle	Ultrasound; aspiration if symptomatic or uncertain
Mastitis/abscess	Pain, erythema, fever, lactation association; abscess is fluctuant/collection	Antibiotics; continue drainage/feeding if lactational; abscess drainage when needed
Intraductal papilloma	Spontaneous unilateral nipple discharge, often bloody/serous	Diagnostic imaging and surgical/pathology evaluation
Breast cancer	Hard fixed mass, skin/nipple changes, nodes; may be screen-detected	Triple assessment: exam, imaging, tissue diagnosis
Inflammatory breast cancer	Rapid erythema, edema, peau d orange, warmth, tenderness, often no discrete mass	Urgent diagnostic workup; do not treat repeatedly as simple mastitis if not resolving

Screening relationship

Asymptomatic average-risk patient -> screening mammography. Symptomatic patient or abnormal exam -> diagnostic imaging and tissue evaluation pathway. A normal screening schedule does not rule out cancer in a patient with a new focal finding.

14B.14 Vulvar, Vaginal, and Lower Genital Tract Disorders

Vulvar and vaginal symptoms often overlap with infection, dermatology, menopause, autoimmune disease, pain syndromes, and malignancy. Pruritus can be candidiasis, lichen sclerosus, contact dermatitis, psoriasis, lichen planus, vulvar intraepithelial neoplasia, or malignancy. Pain can be infectious, inflammatory, neuropathic, pelvic floor-related, traumatic, or hormonal. Persistent vulvar symptoms deserve examination; chronic empiric antifungal treatment without reassessment can delay diagnosis of dermatoses or cancer.

Lichen sclerosus classically causes porcelain-white atrophic plaques, severe pruritus, fissures, architectural changes, and dyspareunia. It is associated with increased vulvar squamous cell carcinoma risk. High-potency topical corticosteroids are the usual treatment foundation, and persistent lesions or suspicious changes require biopsy. Lichen planus can involve the vulva and vagina and may cause erosive disease, discharge, pain, and scarring.

Disorder	Clues	Why it matters
Contact dermatitis	Itching/burning after irritants, pads, soaps, topical products	Remove trigger; avoid mislabeling as recurrent yeast
Lichen sclerosus	White atrophic plaques, pruritus, fissures, architectural changes	Vulvar SCC risk; biopsy suspicious lesions; high-potency steroid treatment
Lichen planus	Painful erosions, Wickham striae, vaginal involvement/scarring	Can cause stenosis and dyspareunia; often needs specialist management
Vulvodynia	Chronic vulvar pain without clear cause after evaluation	Pelvic floor, neuropathic, behavioral, and multidisciplinary care may be needed
Vulvar cancer	Persistent ulcer, mass, color change, bleeding, pruritus, nonhealing lesion	Biopsy persistent or suspicious lesions

14B.15 Integrated Diagnostic Algorithms

Algorithm A: abnormal bleeding with reproductive disease context

Step 1: assess hemodynamic stability. Heavy bleeding with syncope, hypotension, severe anemia symptoms, or pregnancy concern requires urgent evaluation. Step 2: obtain pregnancy testing in reproductive-age patients. Step 3: classify by age and context: adolescent anovulation/coagulopathy, reproductive-age pregnancy/anovulation/structural disease, perimenopausal ovulatory dysfunction plus malignancy risk, or postmenopausal bleeding. Step 4: use PALM-COEIN logic: structural causes include polyp, adenomyosis, leiomyoma, malignancy/hyperplasia; nonstructural causes include coagulopathy, ovulatory dysfunction, endometrial, iatrogenic, and not otherwise classified. Step 5: decide whether endometrial sampling, pelvic ultrasound, labs, or acute bleeding therapy is needed.

Algorithm B: pelvic pain with infection and malignancy forks

First exclude pregnancy-related emergencies and hemodynamic instability. Then sort pain into acute vs chronic, cyclic vs noncyclic, infectious vs noninfectious, and mass/torsion vs inflammatory. Fever, discharge, cervical motion tenderness, uterine/adnexal tenderness, or STI risk should raise PID concern. Sudden severe unilateral pain with nausea/vomiting raises torsion concern. Cyclic pain with dyspareunia or dyschezia suggests endometriosis. Bloating, early satiety, ascites, or postmenopausal mass symptoms raise ovarian malignancy concern.

Algorithm C: discharge

Determine whether the discharge is vaginal, cervical, urinary, or physiologic. Vaginal itching and odor suggest vaginitis. Cervical friability, mucopurulent discharge, postcoital bleeding, or STI risk suggests cervicitis. Pelvic pain or tenderness suggests PID. Check pH/wet mount/NAATs based on syndrome. Treat partners when STI is diagnosed or strongly suspected, and consider HIV/syphilis testing where indicated.

Algorithm D: postmenopausal bleeding

Any bleeding after menopause deserves evaluation. Common causes include atrophy, polyps, anticoagulation, hyperplasia, and cancer, but the key principle is that benign causes are diagnoses after evaluation. Transvaginal ultrasound and/or endometrial sampling are used depending on bleeding pattern, endometrial thickness, risk factors, and persistence. Recurrent bleeding requires continued evaluation even after an initially reassuring test.

14B.16 Graph-RAG Relationship Summary

The following relationship statements are deliberately redundant and extraction-friendly. They are phrased as concept-to-concept links that should be useful for graph construction.

- PCOS -> chronic anovulation -> irregular menses and infertility.
- PCOS -> insulin resistance -> hyperinsulinemia -> increased ovarian androgen production.
- Hyperandrogenism -> hirsutism, acne, androgenic alopecia, and ovulatory dysfunction.
- Chronic anovulation -> unopposed estrogen -> endometrial hyperplasia -> endometrial cancer risk.
- Combined hormonal contraception -> ovarian suppression and endometrial stabilization -> cycle control in PCOS.
- Spironolactone -> androgen receptor blockade -> reduced hirsutism but requires contraception when pregnancy is possible.
- Endometriosis -> ectopic endometrial-like tissue -> inflammation, adhesions, cyclic pain, dyspareunia, dyschezia, and infertility.
- Adenomyosis -> endometrial tissue in myometrium -> enlarged tender uterus, dysmenorrhea, and heavy bleeding.
- Submucosal fibroid -> endometrial cavity distortion -> heavy bleeding and infertility.
- Heavy menstrual bleeding -> iron deficiency anemia -> fatigue, dyspnea, palpitations, and reduced exercise tolerance.
- Estrogen-containing contraception -> increased VTE risk in susceptible patients -> avoid with migraine with aura, prior VTE/stroke, severe hypertension, or high-risk thrombophilia.
- Levonorgestrel IUD -> local endometrial suppression -> decreased heavy menstrual bleeding and endometrial protection.
- Copper IUD -> local spermicidal inflammatory effect -> hormone-free contraception but can worsen bleeding/cramping.
- Bacterial vaginosis -> loss of lactobacilli and anaerobic overgrowth -> thin discharge, fishy odor, high vaginal pH, clue cells.
- Candida vaginitis -> yeast overgrowth -> vulvar pruritus, burning, erythema, and usually normal vaginal pH.
- Trichomoniasis -> sexually transmitted protozoan infection -> discharge, irritation, elevated pH, partner treatment need.
- Cervicitis -> chlamydia/gonorrhea risk -> mucopurulent discharge and cervical friability.
- Untreated cervicitis -> ascending infection -> PID.
- PID -> salpingitis and tubal scarring -> infertility, ectopic pregnancy, and chronic pelvic pain.
- PID -> tubo-ovarian abscess -> sepsis/rupture risk and hospitalization need.
- Persistent high-risk HPV -> cervical dysplasia -> cervical cancer.
- HPV vaccination -> reduced HPV-associated cancer and wart risk but does not replace screening.
- Abnormal cervical screening -> risk-based triage -> colposcopy and biopsy when indicated.
- Postmenopausal bleeding -> endometrial cancer evaluation until proven otherwise.
- Obesity -> adipose aromatization -> increased estrogen exposure -> endometrial cancer risk.
- BRCA1/2 pathogenic variants -> hereditary breast and ovarian cancer risk -> genetic counseling and risk-reduction planning.
- Lynch syndrome -> mismatch repair defect -> colorectal, endometrial, and ovarian cancer risk.
- Ovarian cancer -> nonspecific bloating/early satiety/abdominal distension -> delayed diagnosis risk.
- CA-125 -> nonspecific tumor marker -> not a population screening test for average-risk ovarian cancer.
- Screening mammography -> asymptomatic average-risk breast cancer detection; diagnostic imaging -> symptomatic breast finding evaluation.
- Inflammatory breast cancer -> rapid erythema/edema/peau d orange -> urgent cancer workup if not resolving as mastitis.
- Lichen sclerosus -> chronic vulvar inflammation and scarring -> vulvar squamous cell carcinoma risk.

14B.17 Review Tables

Diagnosis	Classic pattern	One-liner treatment/next step
PCOS	Oligo/anovulation + hyperandrogenism +/- polycystic ovaries; metabolic risk	Exclude mimics; protect endometrium; treat hyperandrogenism/metabolic risk/fertility goals
Endometriosis	Cyclic pelvic pain, deep dyspareunia, dyschezia, infertility	NSAIDs/hormonal suppression; surgery if refractory, mass, or fertility indications
Adenomyosis	Heavy painful periods, bulky tender uterus	NSAIDs, hormonal suppression, LNG-IUD; hysterectomy definitive if no fertility desire
Fibroids	Bleeding or bulk symptoms depending location	TVUS; medical bleeding control or procedure based on symptoms/fertility
BV	Fishy odor, thin discharge, pH >4.5, clue cells	Metronidazole/clindamycin regimens
Candida	Vulvar pruritus/burning, thick discharge possible, normal pH	Azole/fluconazole; longer therapy if complicated/recurrent
Trichomoniasis	STI vaginitis with elevated pH, discharge/irritation	Nitroimidazole treatment and partner treatment
PID	Pelvic pain + CMT/uterine/adnexal tenderness	Empiric broad antibiotics; hospitalize high-risk/severe/TOA/pregnant/no response
Cervical dysplasia	Abnormal Pap/HPV screening	Risk-based triage; colposcopy/biopsy/treatment when indicated
Endometrial cancer	Postmenopausal bleeding; unopposed estrogen risks	TVUS/endometrial sampling pathway
Ovarian cancer	Bloating, early satiety, pelvic/abdominal pain, adnexal mass	Risk stratify mass; specialist referral when concerning

References and Guideline Anchors

1. Centers for Disease Control and Prevention. Sexually Transmitted Infections Treatment Guidelines, 2021. CDC; updated online guidance accessed 2026.
2. Centers for Disease Control and Prevention. U.S. Medical Eligibility Criteria for Contraceptive Use, 2024. MMWR Recomm Rep. 2024;73(4):1-126.
3. U.S. Preventive Services Task Force. Cervical Cancer: Screening. Recommendation statement and draft update materials for HPV-based screening strategies.
4. Women's Preventive Services Initiative / HRSA. Women's Preventive Services Guidelines: cervical cancer screening and preventive services.
5. U.S. Preventive Services Task Force. Breast Cancer: Screening. Final Recommendation Statement. 2024.
6. International PCOS Network and ASRM. Recommendations from the 2023 International Evidence-Based Guideline for the Assessment and Management of Polycystic Ovary Syndrome.
7. American College of Obstetricians and Gynecologists. Hereditary Cancer Syndromes and Risk Assessment. Committee Opinion. 2019.
8. American College of Obstetricians and Gynecologists. Hereditary Breast and Ovarian Cancer Syndrome. Practice Bulletin No. 182. 2017.
9. American College of Obstetricians and Gynecologists. Lynch Syndrome. Practice Bulletin No. 147. 2014.
10. National Cancer Institute. Cervical Cancer Screening. Updated patient and clinician-facing screening overview.

Use note

This chapter is an original educational synthesis for internal medicine study and graph-RAG ingestion. It is not a substitute for local protocols, specialist guidance, or patient-specific clinical judgment. Screening and treatment recommendations can change; use current CDC, USPSTF, HRSA/WPSI, ACOG, ASCCP, NCCN, and institutional guidance for clinical decisions.